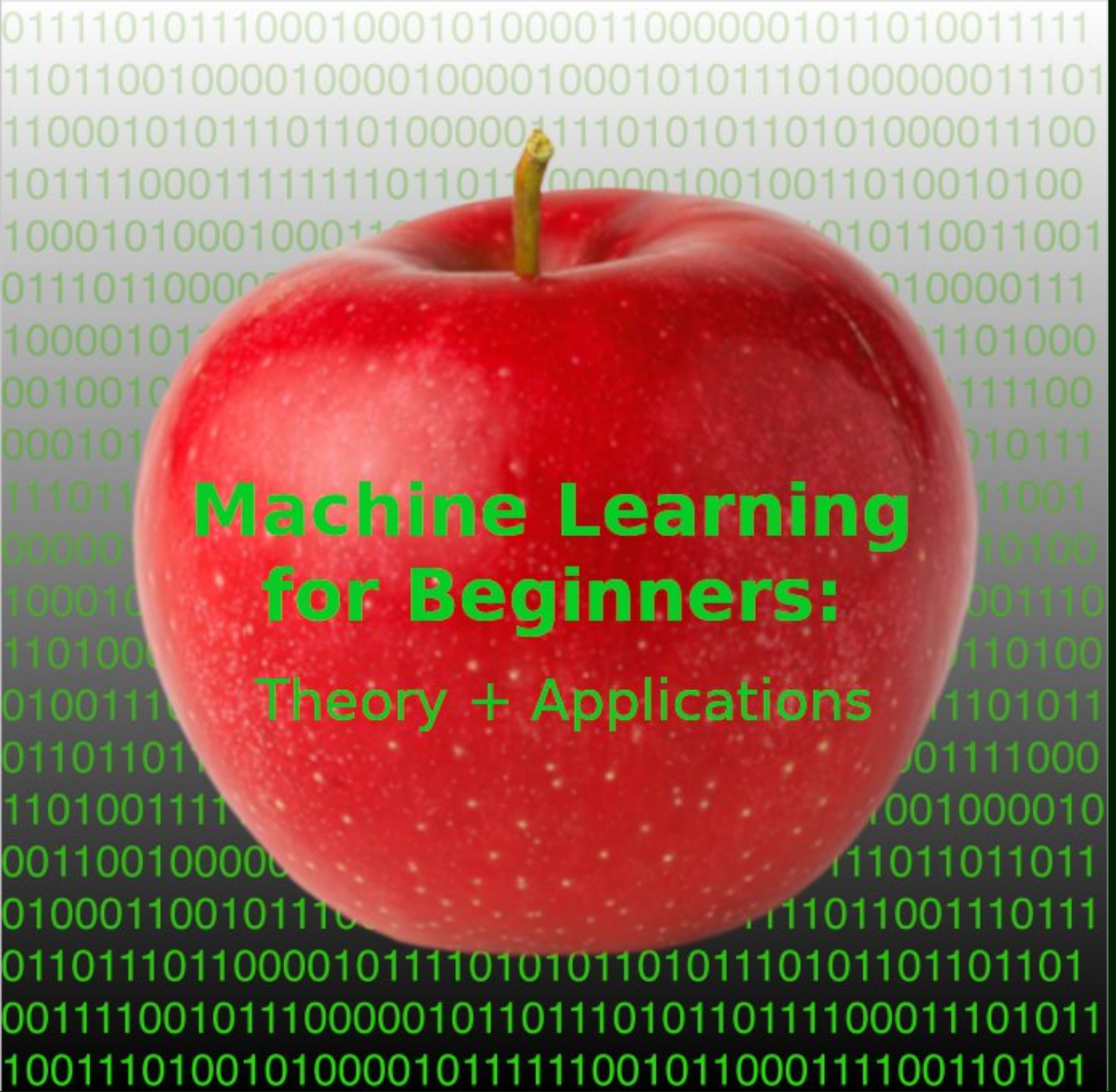


University of Bayreuth
Chair for Polymer Engineering
Prof. Dr. Holger Ruckdäschel

Title:

"Machine Learning for Beginners: Theory and Applications"

(Available in E-learning/Moodle)

Lecturer:

Dr. Rodrigo Q. Albuquerque

Lecture 10

Topics

Week 1: Introduction
ML types, data, ethics, case studies

Week 2: Preprocessing & *k*NN
Preprocessing, *k*-nearest neighbors, distance metrics, hyperparameters, cross-validation

Week 3: Linear Regression
Types, loss function, (over/under)fitting bias/variance, L1/L2 regularization, metrics

Week 4: Classification
Logistic regression, SVM, classification evaluation metrics, confusion matrix one-hot/label encoding

Week 5: Random forests
Decision trees, random forests, classification and regression applications

Week 6: PCA / KMeans
Dimensionality reduction, clustering

Week 7: Extra exercises

Week 8: Simple ANNs
Feedforward networks, activation functions, basic training

Week 9: Gaussian Processes
Kernels, gaussian processes

★ **Week 10: Design of Experiments**
DoE, random design, active learning, bayesian optimization

Weeks 11-12: Student projects

Design of Experiments (DoE)

Factorial design

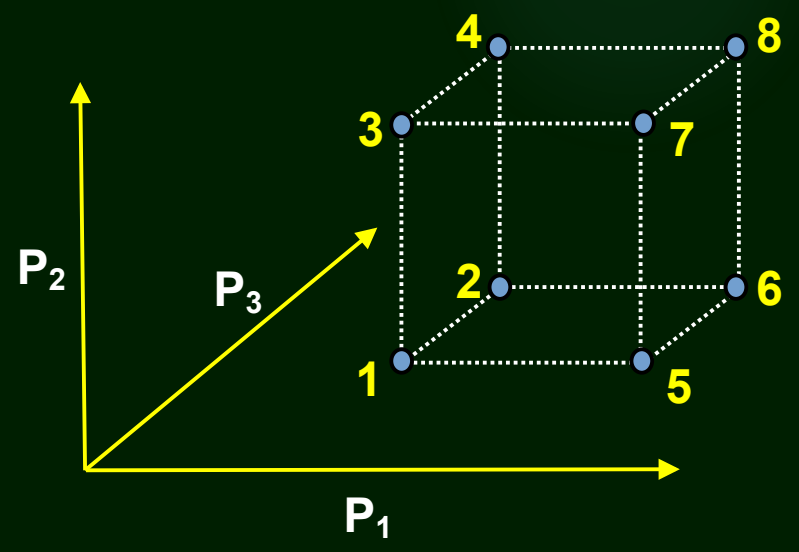
3 Parameters: P_1, P_2, P_3

2 Levels:
P_1: 10 or 30
P_2: 0.1 or 0.3
P_3: 4 or 6

Exps = 2^3 = 8

Exp #	P_1	P_2	P_3	Target
1	10	0.1	4	145
2	10	0.1	6	136
3	10	0.3	4	151
4	10	0.3	6	139
5	30	0.1	4	142
6	30	0.1	6	148
7	30	0.3	4	133
8	30	0.3	6	137

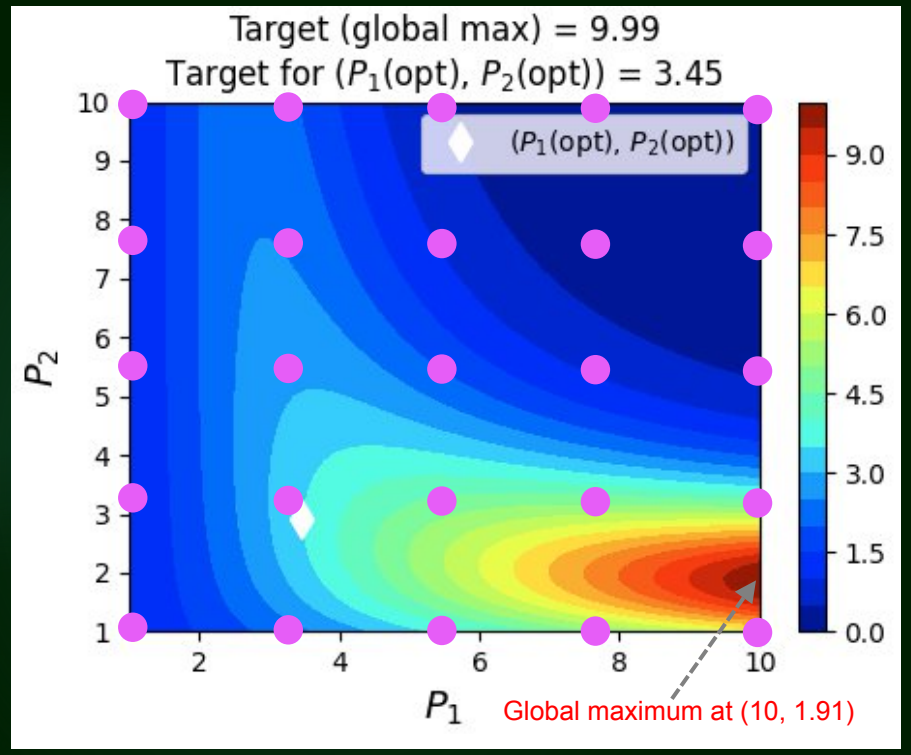
Going to high dimensions:
8 parameters, 5 levels
Exps = 5^8 = 390625 (!!!)



Optimizing parameters

$$target = P_1^{\sin\left(\frac{P_1 P_2}{P_1 + P_2}\right)}$$

DoE



2 Parameters: P_1, P_2

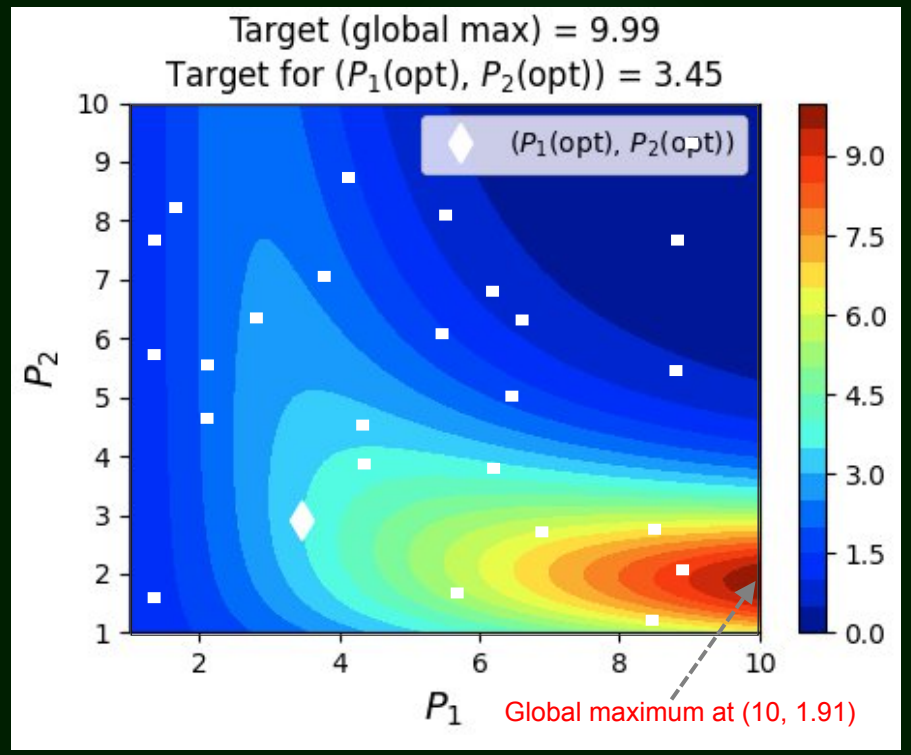
5 Levels:

P_1: 1, 3.25, 5.50, 7.75, 10

P_2: 1, 3.25, 5.50, 7.75, 10

Exps = 5^2 = 25

Random Design



After 25 measurements, the best target found would be 6.15 for P_1 = 10 and P_2 = 1

On average (1000 repetitions), after 25 random measurements, the best target = 7.62 +/- 1.35

Active Learning (AL)

1) Select random experiments

("Random Design")

2) Perform the experiments

3) Train a model (Gaussian Processes)

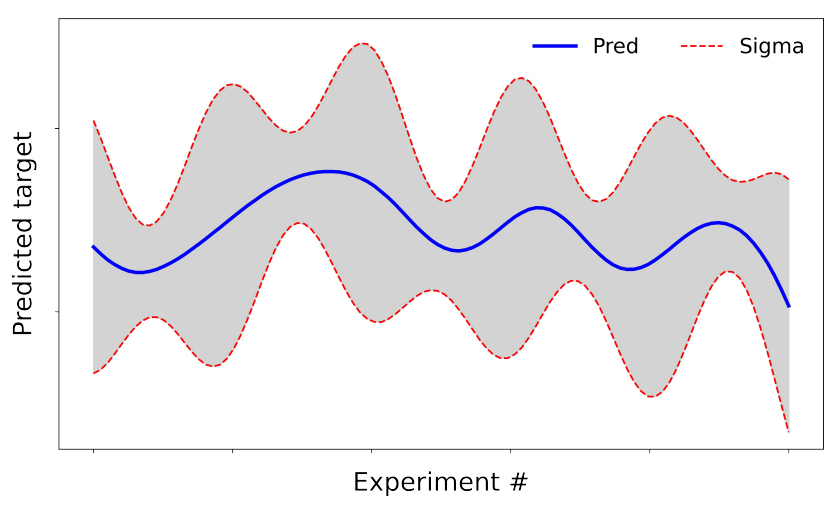
Predictions (10^6 virtual experiments)

Exp #	P_1	P_2	P_3	P_4	P_5	P_6	Target
1	10	0.1	3	9	27	1.5	132
2	12	0.5	1	7	24	1.9	155
3	11	0.4	2	6	25	1.6	143
4	10	0.3	1	8	26	1.8	145

Loop to:

i) Grow the dataset

ii) Improve the model



virtual experiments

#	Pred	Sigma
1	-	-
2	-	-
⋮		
10^6	-	-

4) Calculate AL property:

i) Entropy

ii) Distance

iii) Uncertainty

iv) Hybrid

5) Get experiment with largest AL property

Bayesian Optimization (BO)

1) Select random experiments

(“Random Design”)

2) Perform the experiments

3) Train a model (Gaussian Processes)

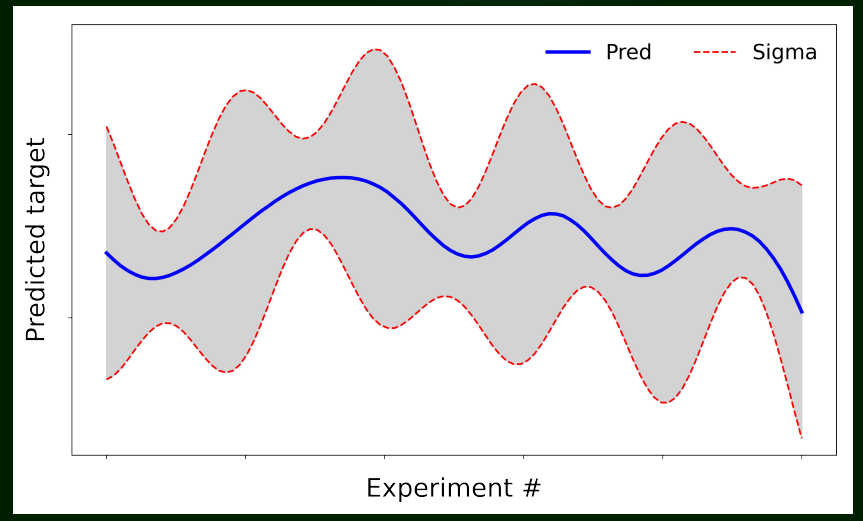
Predictions
(10^6 virtual experiments)

Exp #	P_1	P_2	P_3	P_4	P_5	P_6	Target
1	10	0.1	3	9	27	1.5	132
2	12	0.5	1	7	24	1.9	155
3	11	0.4	2	6	25	1.6	143
4	10	0.3	1	8	26	1.8	161

Loop until:

i) Target is high enough

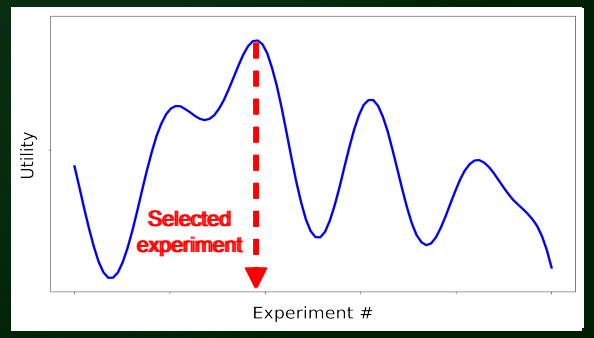
ii) Budget is over



virtual experiments

#	Pred	Sigma
1	-	-
2	-	-
⋮		
10^6	-	-

4) Acquisition function



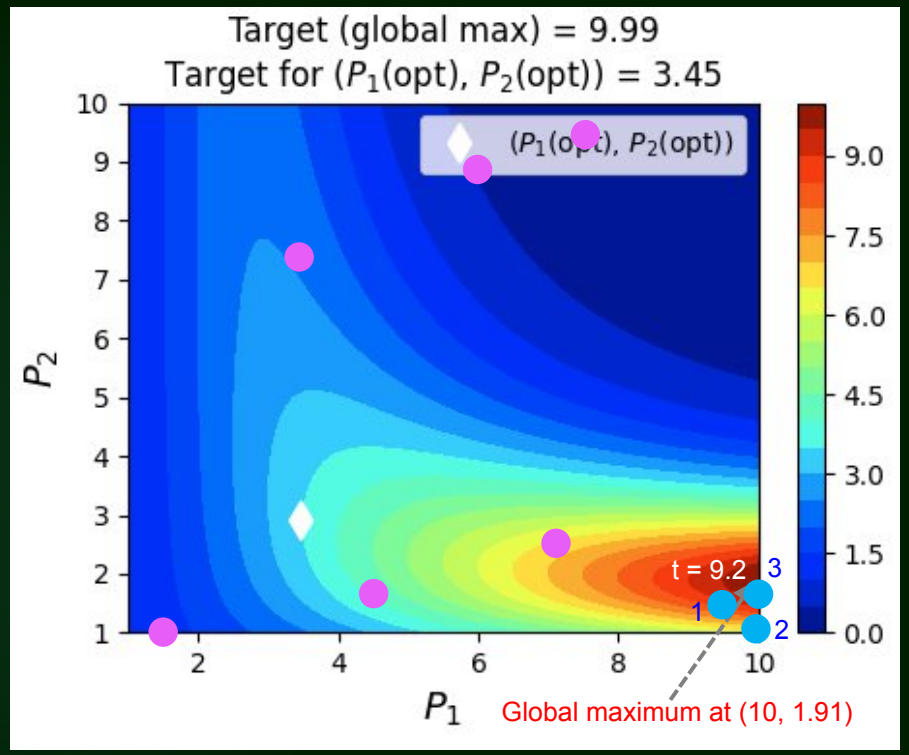
https://mms.uni-bayreuth.de/Panopto/Pages/Viewer.aspx?id=73e38829-ce7d-4fee-8f61-b341008e2723

Optimizing parameters

$$target = P_1^{\sin\left(\frac{P_1 P_2}{P_1 + P_2}\right)}$$

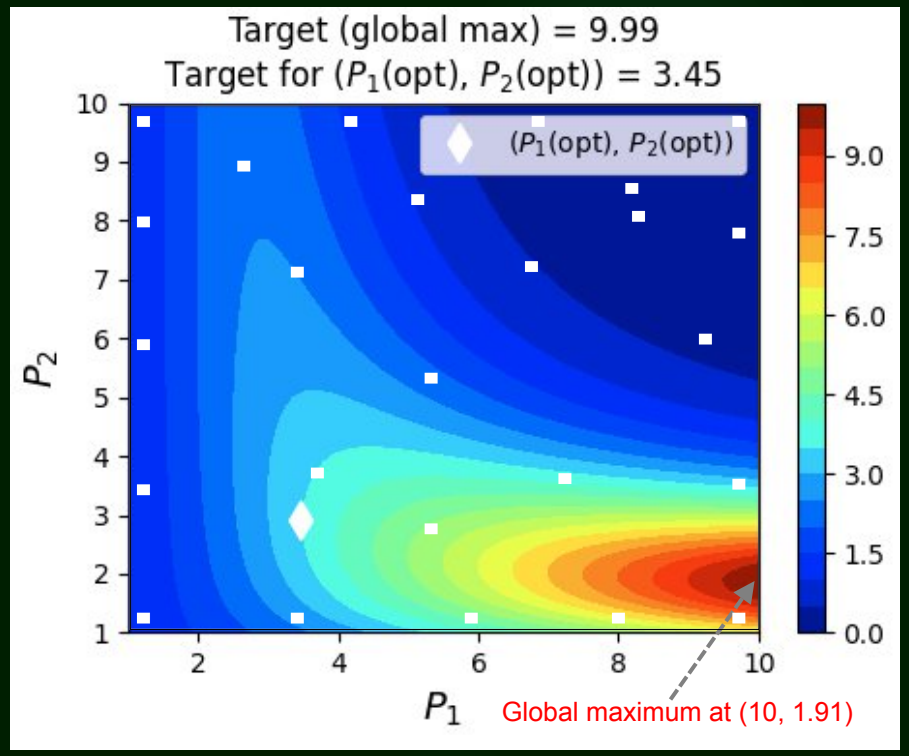
Note:
Manual BO is much more efficient!
(The BO statistics shown was for automatic BO)

Random Design + BO



For 30 repetitions (for reaching t > 9):
exps = 6 Rdn + (7.89 BOs +/- 3.75)

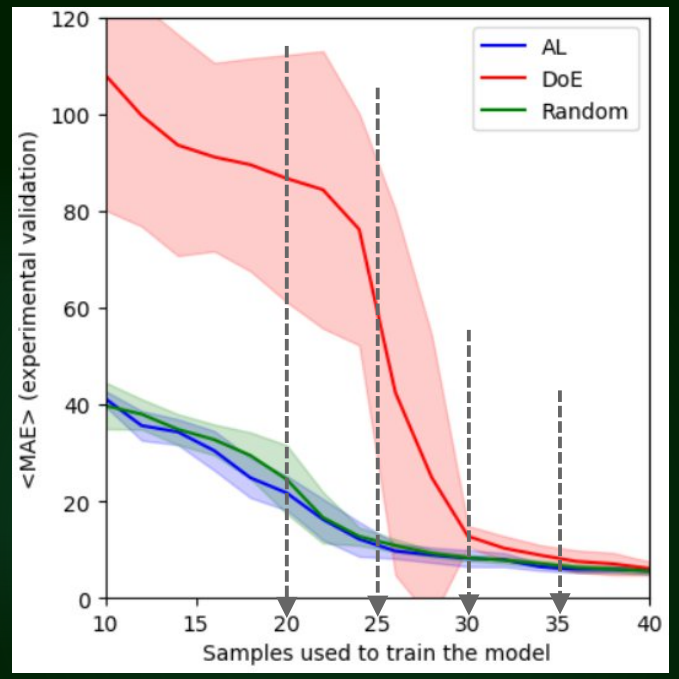
AL



AL explores unknown regions to improve the model

AL x DoE x Rdn

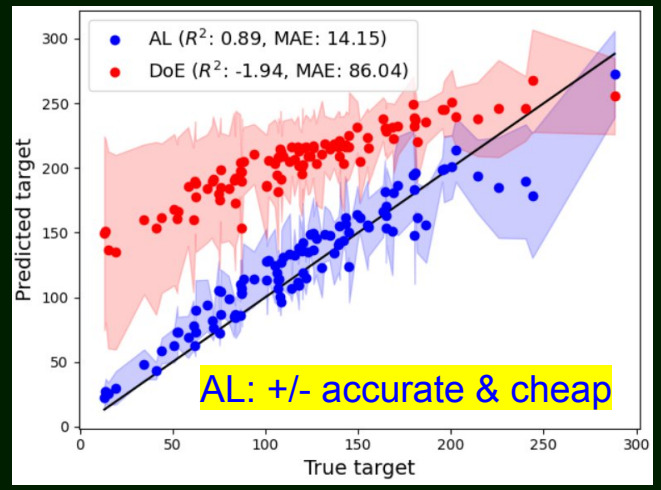
4 parameters, 3 levels
target = Griewank



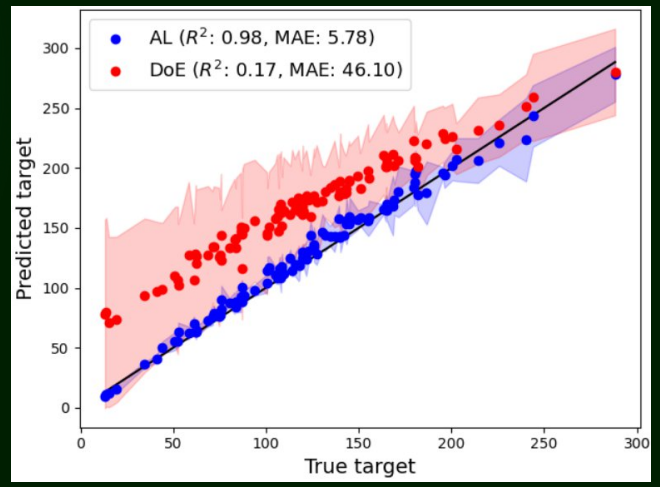
Predictions ⟷ Error

Exp Val
(target)

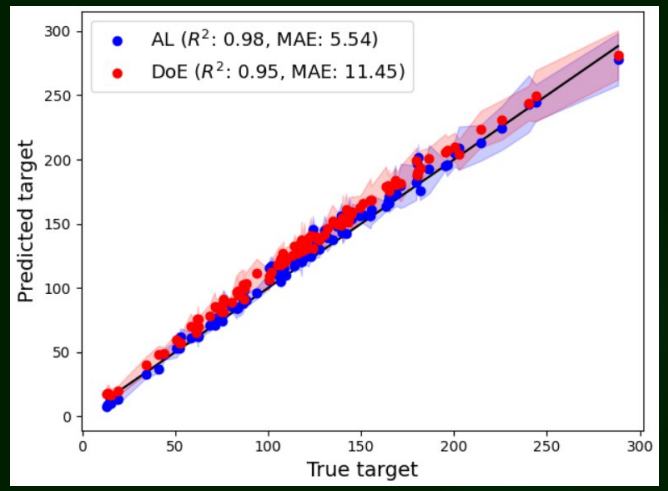
ML trained with 20 samples



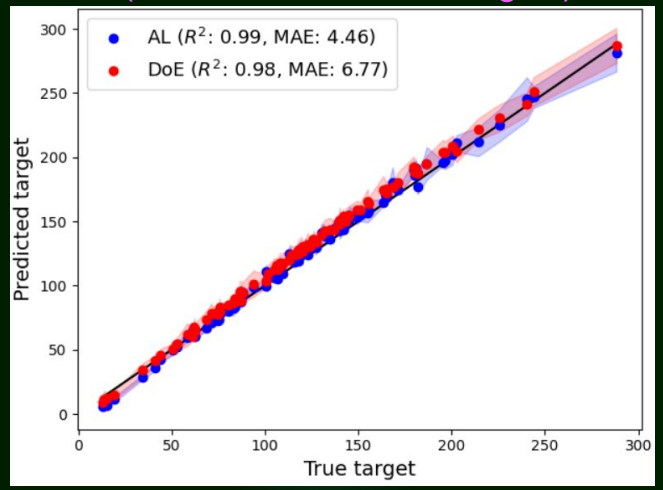
ML trained with 25 samples



ML trained with 30 samples
(DoE error still > 2 AL error!)

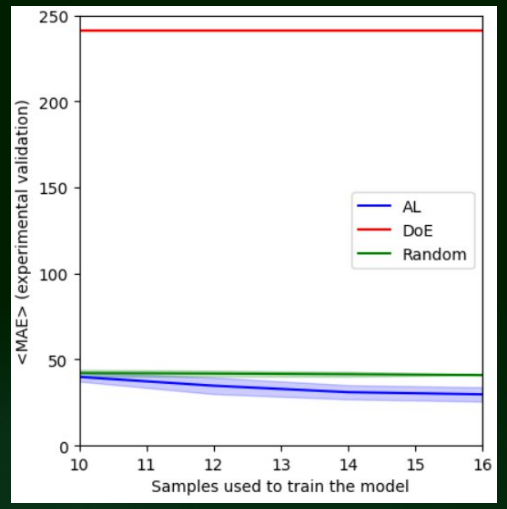


ML trained with 35 samples
(DoE error still 52% larger!)

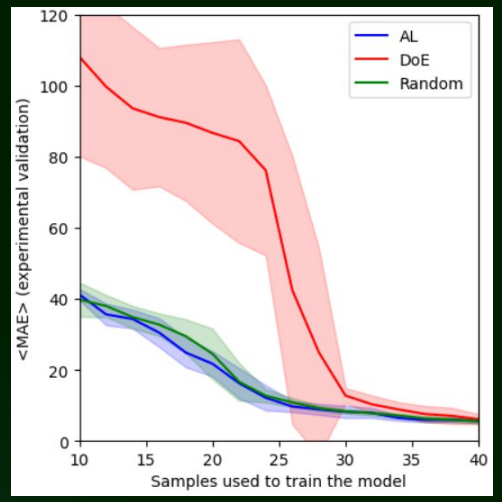


AL x DoE x Rdn

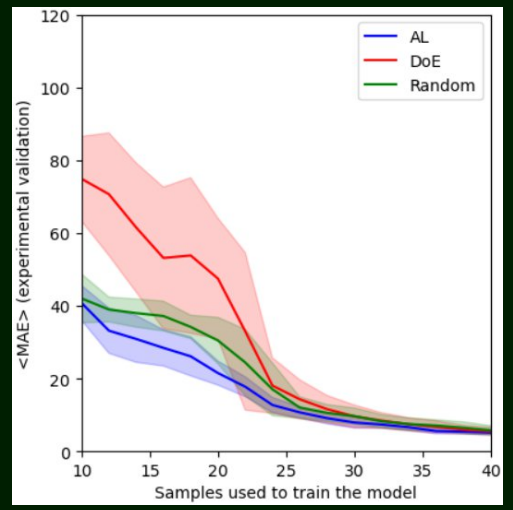
2 levels

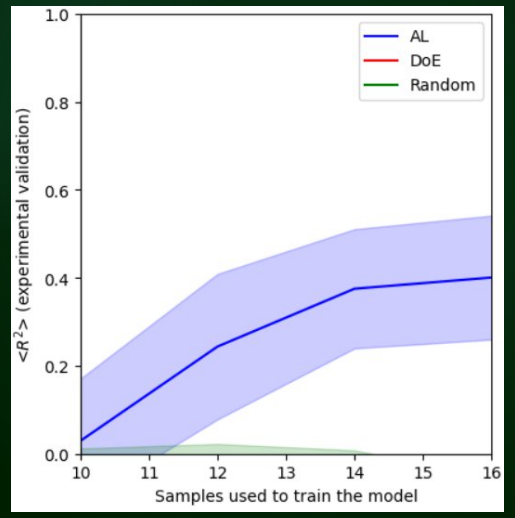


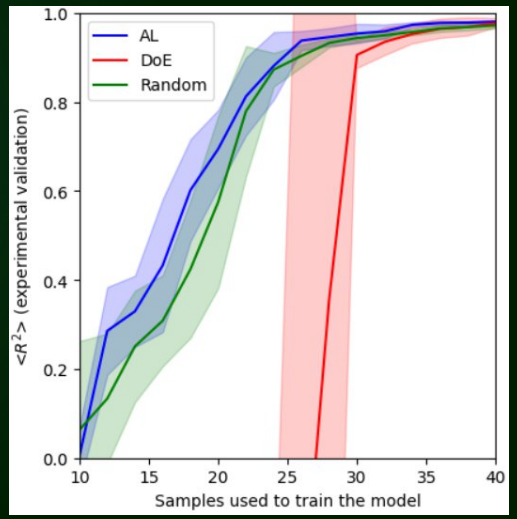
3 levels

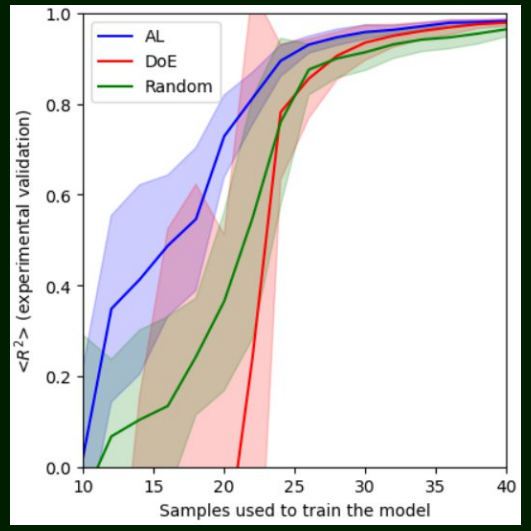


4 levels









4 parameters
target = Griewank

Predictions ⟷ Error

Exp Val
(target)

- DoE improves
- AL better than RDN and DoE

AL x DoE x Rdn

Active learning and Gaussian processes for the development of dissolution models: An AI-based data-efficient approach

Roshan A. Patel **, Siddharth S. Kesharwani, Fady Ibrahim *

Drug Product Development, Synthetics Platform, Sanofi, 350 Water St., Cambridge, MA 02141, USA

Journal of Controlled Release 379 (2025) 316–326 (Link of the paper)

Conclusions of this paper:

AL could have reduced the total number of experiments between 35 % and 50 % to train a model with predictive capabilities equivalent to those attained by a model trained using nearly the entire set of DoE experiments.

We recommend adoption of AL over DoE to avoid redundant experimentation

BO Case Studies

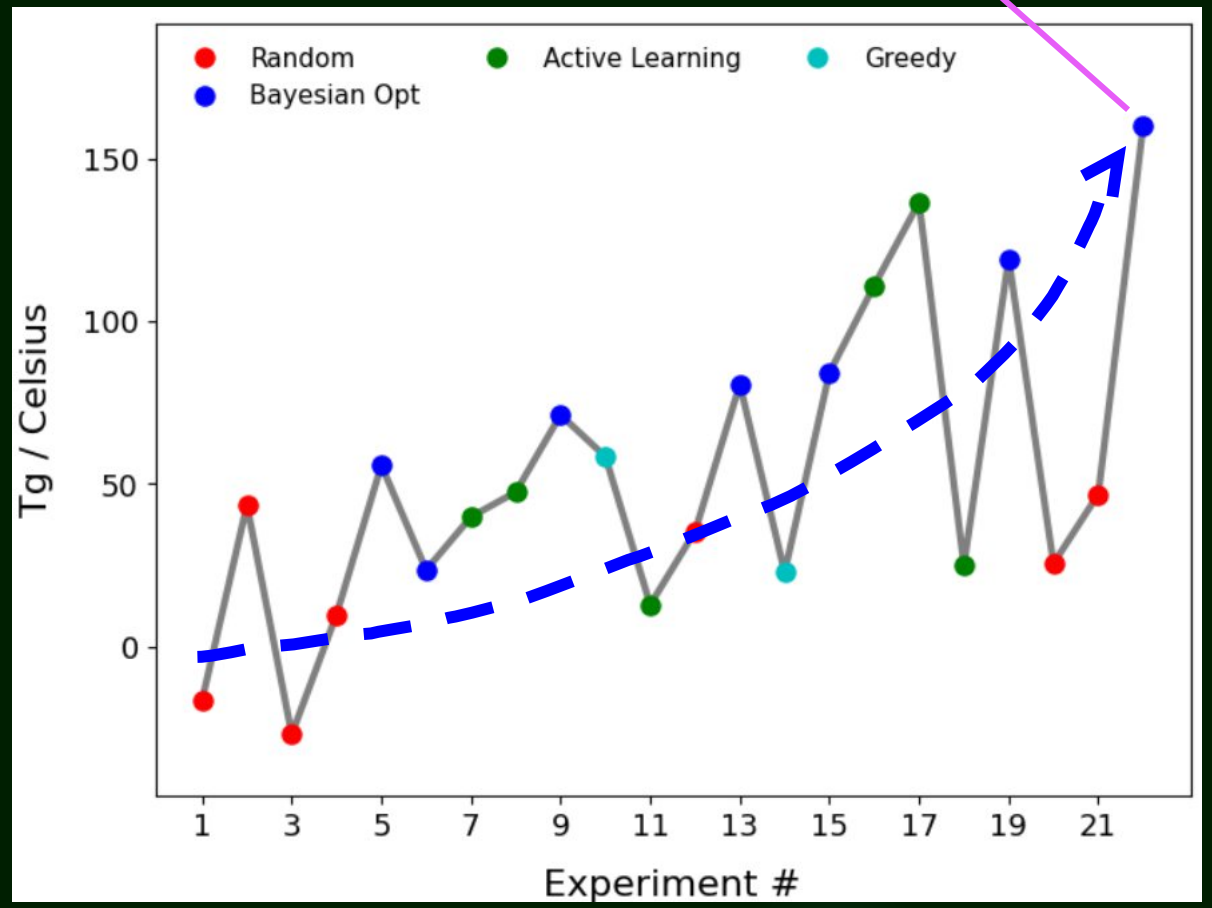
Optimizing a 13-component system

(7 resins + 6 hardeners)

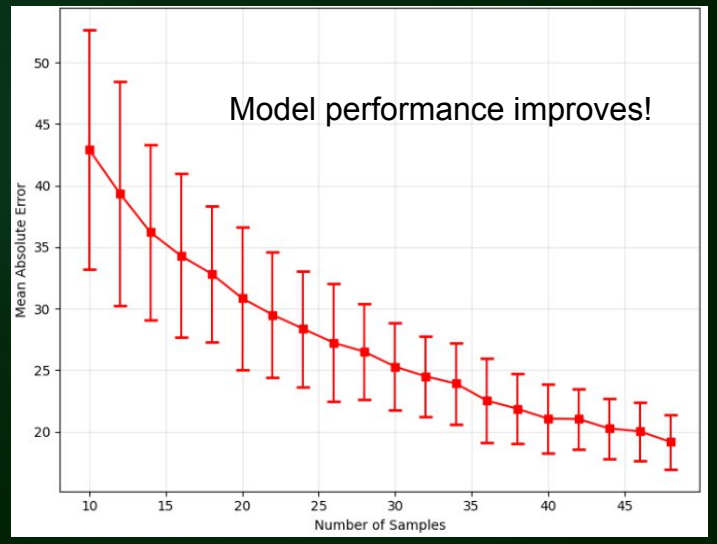
Goal 1: Maximize Tg
Goal 2: Improve ML model to calculate:
- Maximum Tg for desired biocontent
- Maximum biocontent for desired Tg

Exp #22: 0% NC514S; 0% LITE547LV; 0.15% PP1402; 71.27% DER330; 9.27% DEN431; 0% LITE2402; 0% XFN1050; 19.29% IPDA; 0% ECC; 0% EpiloxF1700; 0% XFN1450; 0.01% JeffamineD230; 0% Merginamid L390

Goal 1



Towards Goal 2

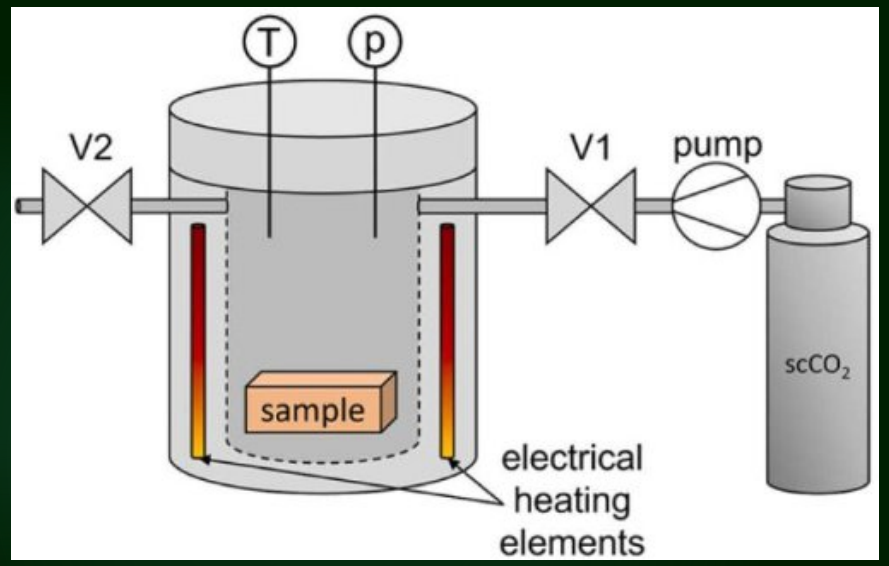


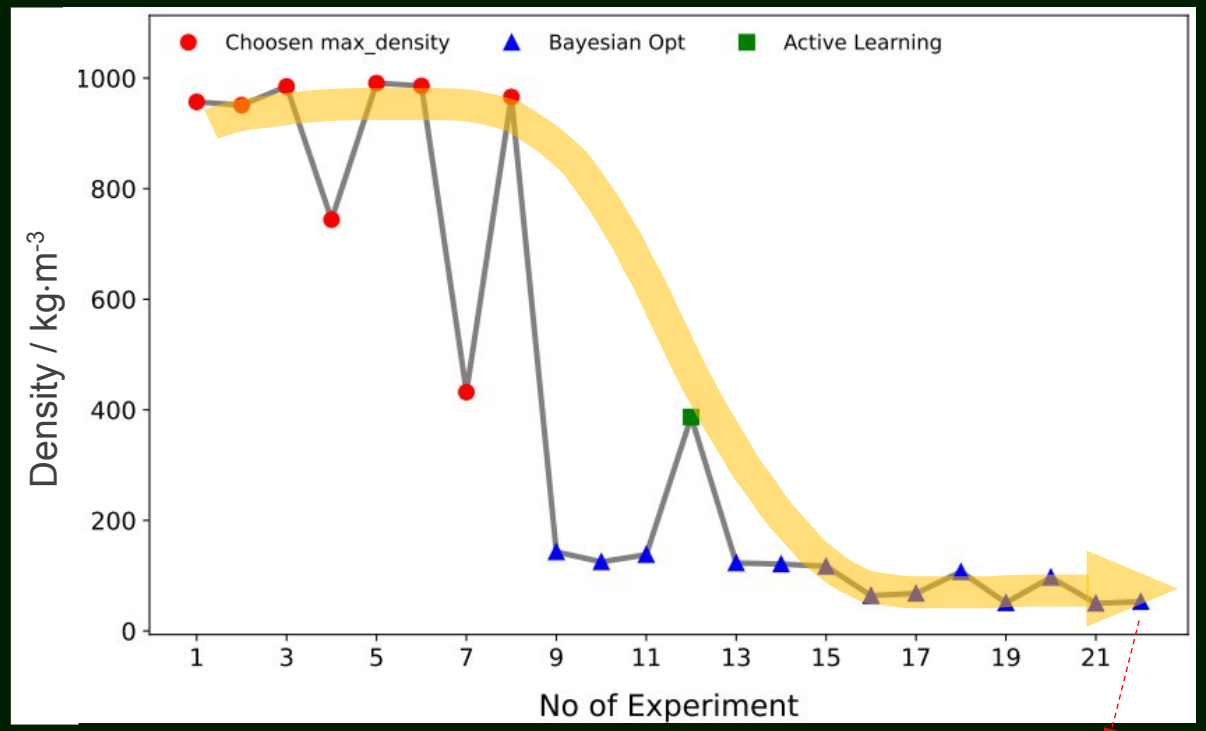
Natalie Wunder et al., paper in preparation.

BO Case Studies

Minimize the density of PA12 foam

- Params: **p, T, time**





(p = 170 bar, T = 188° C, time = 20 min) 50 kg/m^3

Shah *et al.*, Polymer 320 (2025) 128096

Exercises

1) Download the file ‘files10.zip’ from moodle/e-learning (folder “Week 10”) and unzip it locally on your computer
2) Go to google colab (online) **or** launch jupyter notebook locally (offline)
3) Open/upload ‘exercise10.ipynb’ and wait for more instructions